

Q:- Find the domain of ftn, boundary pts, interior pts, determine if the domain is open or closed, bounded or unbounded. Sketch the domain & also shade the region.

① $f(x,y) = \sqrt{16-x^2-y^2} + \sqrt{x^2+y^2-1}$

Sol $16-x^2-y^2 \geq 0$
 $x^2+y^2 \leq 16$

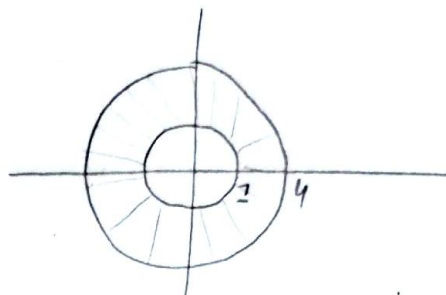
$x^2+y^2-1 \geq 0$
 $x^2+y^2 \geq 1$

Domain = $\{(x,y) \in \mathbb{R}^2 \mid 1 \leq x^2+y^2 \leq 16\}$

The region is closed because it contains all of its boundary pts. The entire region fits inside a disk so it is bounded.

Interior pts are all lying inside the region $\{(x,y) \mid 1 < x^2+y^2 < 4\}$

Boundary pts = = on the circle of radius 1 & 4.
 $x^2+y^2 = 1$ & $x^2+y^2 = 16$



② $f(x,y) = \sqrt{16-x^2-y^2} + \ln(x^2+y^2-4)$

Sol $16-x^2-y^2 \geq 0$

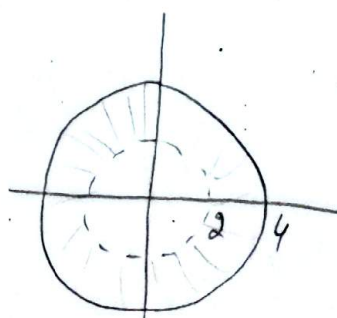
$x^2+y^2 \leq 16$

$x^2+y^2-4 > 0$

$x^2+y^2 > 4$

Domain = $\{(x,y) \in \mathbb{R}^2 \mid 4 < x^2+y^2 \leq 16\}$

This region is neither open nor closed because inner boundary is excluded & outer boundary is included. The region is bounded.



Q3:- Show that ~~e~~ f has partial derivatives at $(0,0)$. Also show that f is not continuous at $(0,0)$. Is f diff at $(0,0)$.

$$f(x,y) = \begin{cases} \frac{xy(1+y^2)}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

Sol

$$f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(0+\Delta x, 0) - f(0,0)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta x \cdot 0 (1+0) - 0}{\Delta x} = 0$$

$$\begin{aligned}
 f(0,0) &= \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0,0)}{\Delta y} \\
 &= \lim_{\Delta y \rightarrow 0} \frac{0 \cdot \Delta y (1 + \Delta y^2) - 0}{\Delta y}
 \end{aligned}$$

$$= 0$$

f has P.D's at $(0,0)$.

For Continuity at $(0,0)$ $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = f(0,0)$

Along $y = mx$

$$\begin{aligned}
 \lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } y=mx}} f(x,y) &= \frac{x \cdot mx (1 + m^2 x^2)}{x^2 + m^2 x^2}
 \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{mx^2 (1 + m^2 x^2)}{x^2 (1 + m^2)} \\
 &= \lim_{x \rightarrow 0} \frac{m (1 + m^2 x^2)}{1 + m^2} \\
 &= \frac{m}{1 + m^2} \quad \text{Path dep}
 \end{aligned}$$

we took Contrapositive statmt
 So f is not Continuous at $(0,0)$ as limit does not exist. f is not Continuous at $(0,0)$ so it is not diff at $(0,0)$.